

Evaluating the Indian Summer Monsoon in the 9 km IFS–FESOM nextGEMS Simulations: Mesoscale Advances, a Persistent Dry Bias, and Its Dynamical Origin



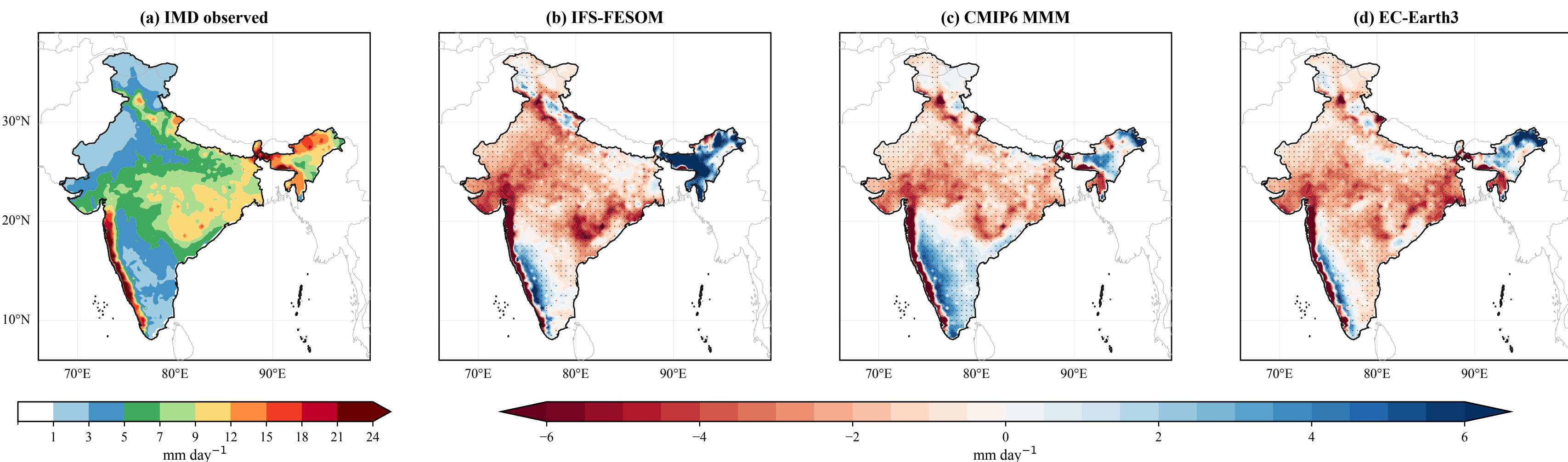
جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

Anirudh Arora^{1,2,3,*}, Hari Prasad Dasari¹, Sanjeev Kumar Jha², Saroj Kanta Mishra³, Ibrahim Hoteit¹

¹ Physical Science & Engineering Div., KAUST, Thuwal, Saudi Arabia ² Department of Earth & Environmental Sciences, IISER Bhopal, India ³ Centre for Atmospheric Sciences, IIT Delhi, India
* Current: Max Planck Institute for Meteorology, Hamburg, Germany *email :anirudh.arora.research@gmail.com

Motivation

The Indian Summer Monsoon is a demanding benchmark for Earth system models: it combines mesoscale convection, orographic forcing, cross-equatorial moisture transport, intraseasonal organization, and ocean-atmosphere feedbacks. ISM delivers ~80% of India's annual rainfall. Yet global models have under-predicted its rainfall across three CMIP generations. Km-scale resolution is a leading hypothesis for closing this gap.



Data & methods

MODEL IFS-FESOM nextGEMS, 9 km atmosphere / 5 km ocean, 1991–2014

REFERENCE IMD 0.25° daily gauge rainfall; ERA5; HadISST v1.1

BASELINE 31 CMIP6 historical models (100–250 km) + multi-model mean

DIAGNOSTICS mean-state & pattern skill; intensity PDFs & Rx1day; orographic stratification; dry-spell / active-break; MISO Hovmöller; SST, Somali & Tropical Easterly jets, vertical shear, IVT

OBJECTIVE

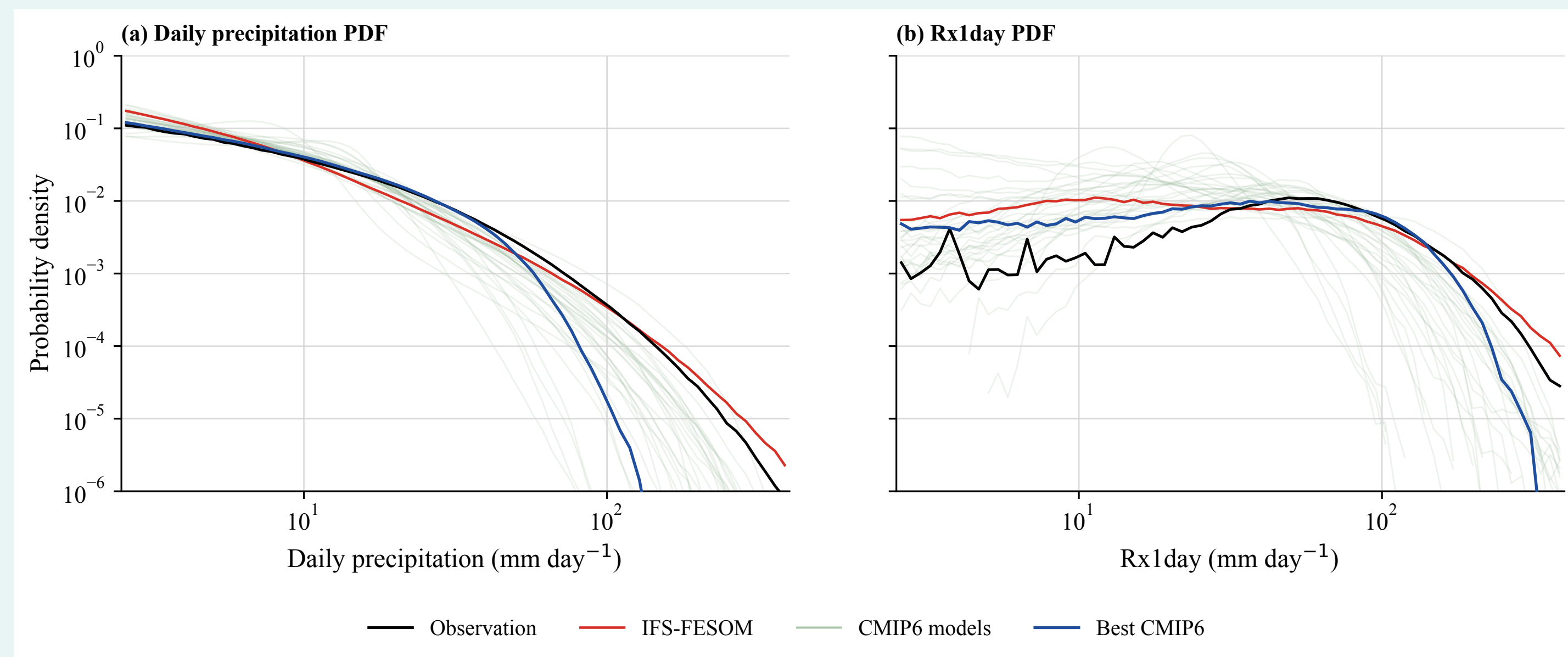
First systematic, process-based evaluation of the Indian monsoon in the IFS-FESOM nextGEMS km-scale simulation; what genuinely improves, and what does not.

The –18% dry bias persists at km-scale and resembles CMIP6, yet IFS-FESOM places rainfall more accurately (PCC 0.691 > 28/31 models).

Where km-scale adds value governed by mesoscale dynamics

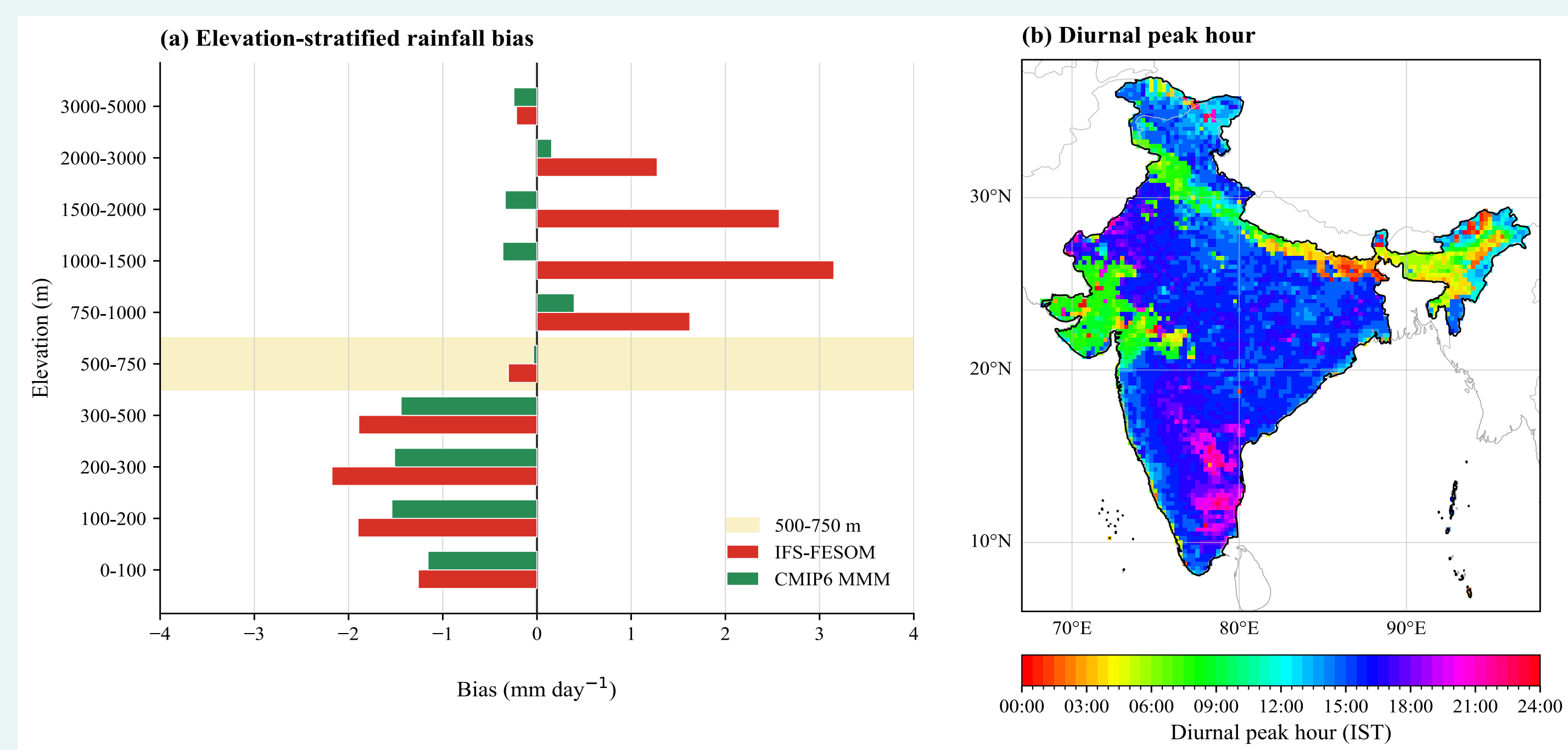
INTENSITY & EXTREMES

Spurious drizzle is suppressed and the heavy tail recovered: median Rx1day reaches **89%** (63 mm d⁻¹) of observed (71 mm d⁻¹) vs **53%** (38 mm d⁻¹) for CMIP6; the 99th percentile (118.5 mm d⁻¹) nearly matches IMD (109.2).



OROGRAPHY & SUB-DAILY TIMING

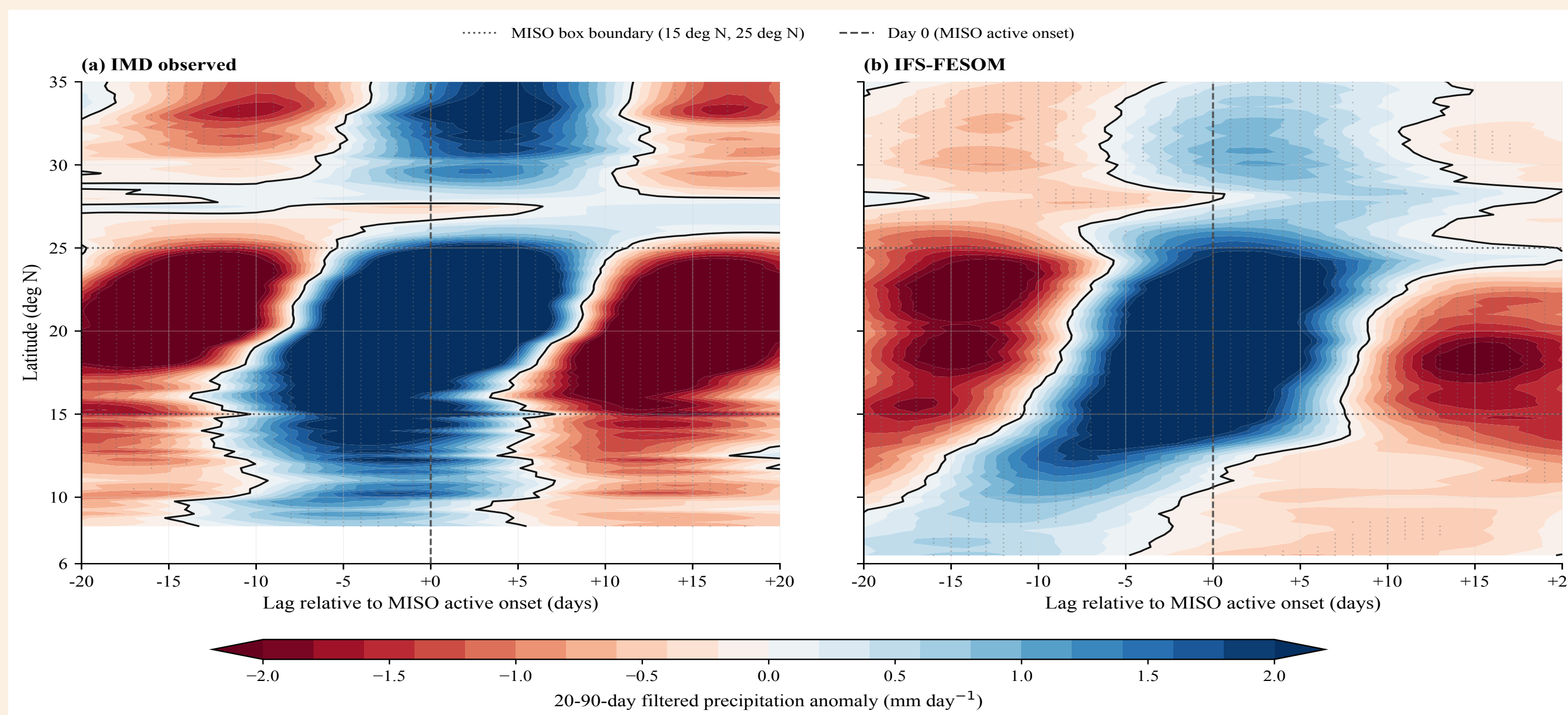
A dry→wet bias crossover at 500–750 m emerges only at km-scale (CMIP6 stays dry), and the afternoon peak (~14:30–15:00 IST) is captured where CMIP6 peaks near noon.



Where biases persist rooted in large-scale coupled dynamics

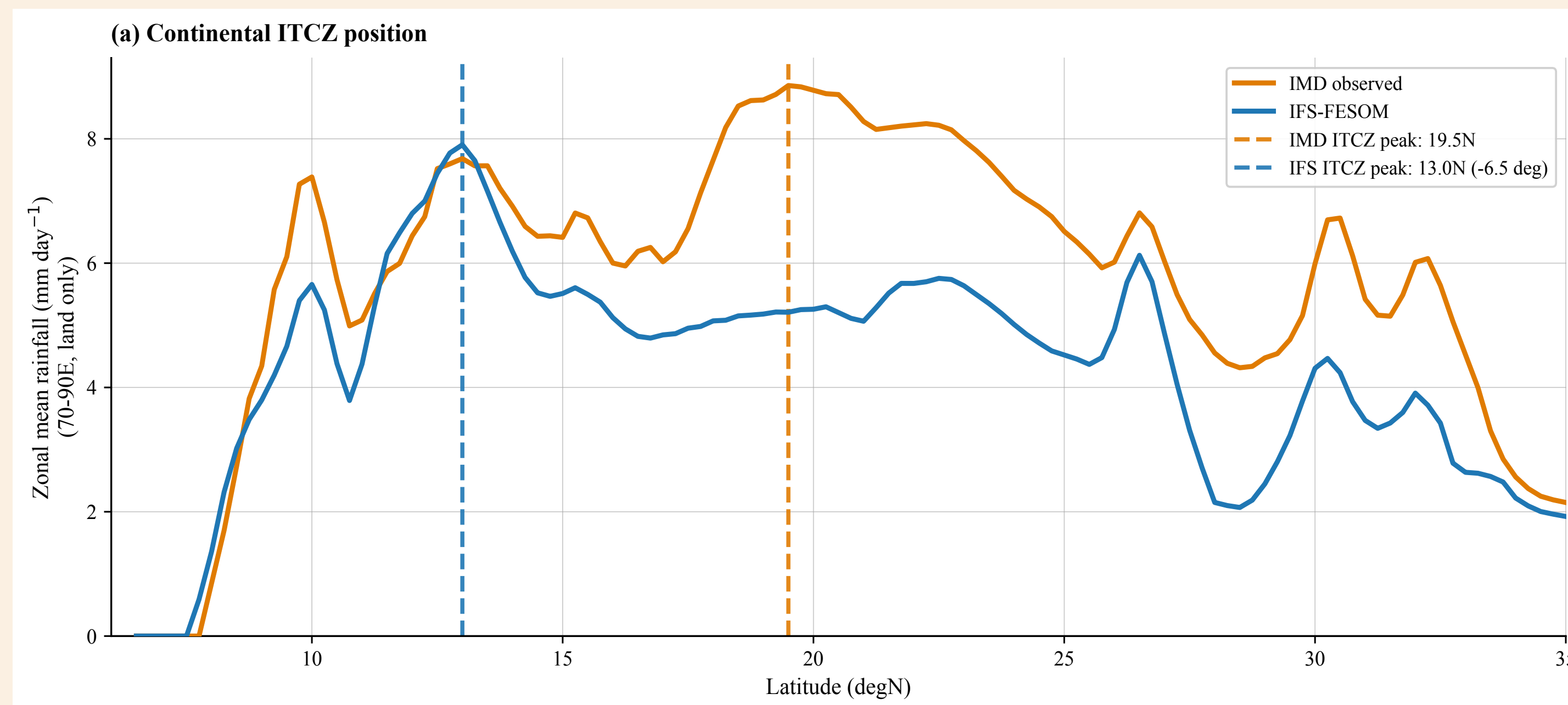
INTRASEASONAL ORGANISATION

The active-break engine fails: observed A-B contrast is **+7.57** mm d⁻¹ but IFS-FESOM gives **–0.45** (wrong sign); dry spells run **~2× too long** (9.78 vs 4.92 d), like CMIP6.

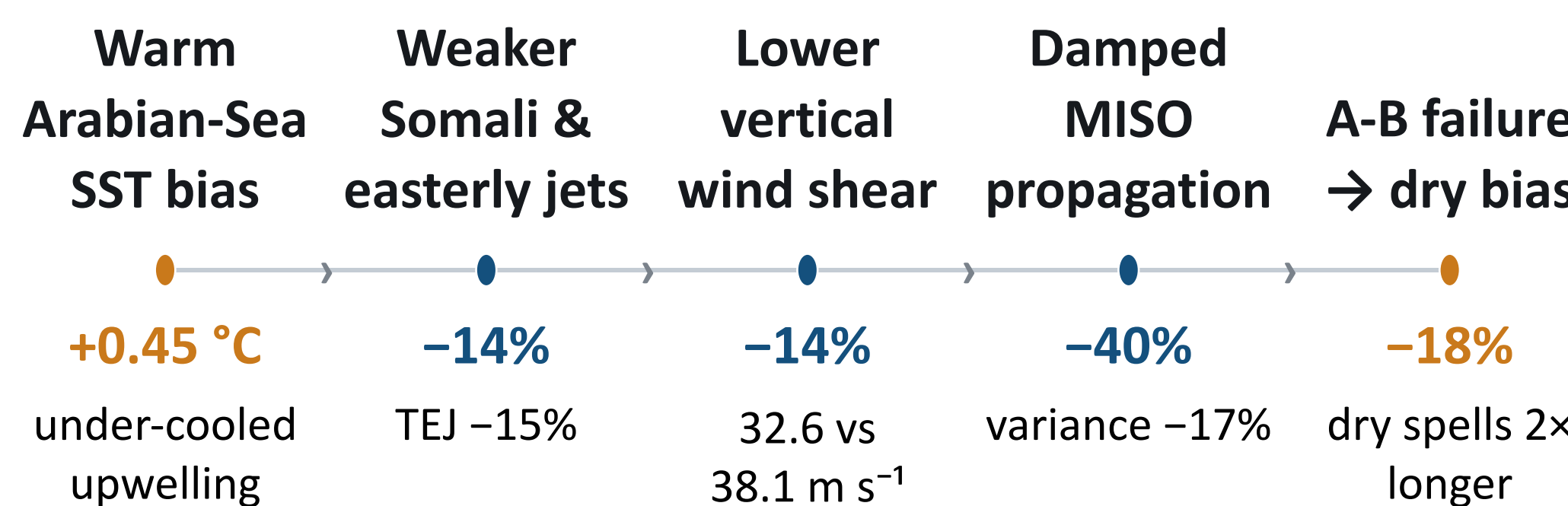
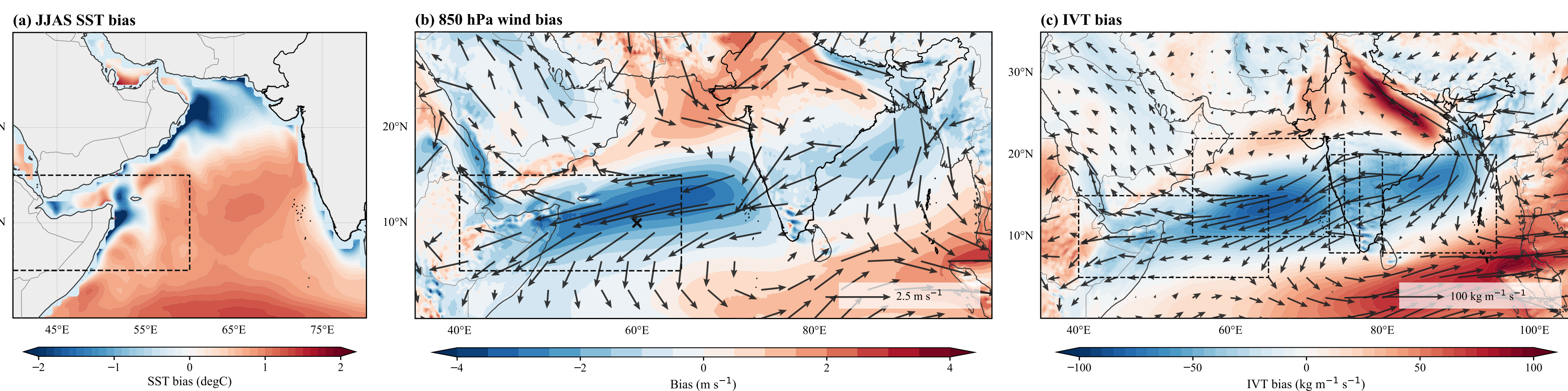


MEAN STATE, ENSO & ONSET

Seasonal amplitude (–18%), interannual skill ($r = +0.15$, ns) and ENSO phasing do not improve; the continental ITCZ is still displaced **–6.5°** equatorward (improved over CMIP6 models). The internal ENSO–ISM bridge, though, is intact ($r = -0.48$).



Why the large-scale biases persist a warm-ocean bias propagates through the circulation into the intraseasonal failure.



Conclusions

- Resolution fixes what it resolves. Drizzle, extreme daily rain (89% vs 53%), orographic gradients and the afternoon peak all improve, from explicitly resolved mesoscale convection and orographic ascent.
- One mechanism explains the failures: a warm Arabian-Sea SST bias → weaker jets → –14% shear → –40% MISO → dry-spell & mean-state bias.

- It does not fix coupled dynamics. Seasonal dry bias, dry-spell length, active-break contrast, MISO amplitude, onset and ENSO skill are no better than CMIP6.
- A clear next test: the intact internal ENSO–ISM bridge ($r = -0.48$) means an AMIP-style prescribed-SST run should recover interannual skill.

SELECTED REFERENCES

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